

Johnny Chan

Zhe Chen | Software, Data Analysis & Technical Operations

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jc8426.github.io/software-data.html | github.com/JC8426

Professional profile

Computer Science PhD researcher combining Python/SQLite application prototyping, structured log analysis and Linux-based systems investigation. Led requirements and AI-assisted implementation of a local request-review workspace. Brings collaborative digital-asset forecasting methodology experience and a practical interest in platform operations, application support and software implementation.

Technical skills

Application development: Python HTTP/JSON APIs, SQLite and SQL queries, HTML/CSS/JavaScript, role-restricted interfaces, signed-session handling, audit records and unit/integration testing.

Data and systems: YAML/JSON parsing, event classification, time-series reconstruction, reconciliation checks, data-quality interpretation, Linux/Ubuntu, Git, C++ and Excel.

Engineering workflow: AI-assisted development with Codex and Claude, requirements definition, code review, reproducible checks, technical documentation and GitHub Pages publishing.

Selected implementation and analysis

Operations Request and Review Workspace

Local prototype | Requirements lead and AI-assisted implementation | 2026

- Implemented a Python HTTP service and browser workflow backed by SQLite, covering request intake, field review, human confirmation, task updates, feedback and status-count summaries.
- Preserved original requests and field provenance; required confirmation before task creation. Added role-restricted endpoints, account management, signed sessions and audit records.
- Used deterministic local rules for field extraction and gap detection. Existing tests cover database behavior, permissions, session tampering and confirmation requirements; all 14 passed an isolated rerun in September 2026.

RCAP GameController Log Analysis

Archived competition logs | New retrospective analysis in September 2026

- Prepared an AI-assisted, reproducible audit of 14 YAML files containing 17,509 timestamped records. Classified custom entry types and reconciled parser counts against the raw-text structure.
- Separated repeated internal-state snapshots from state changes and produced event-composition charts and a state timeline. Distinguished robot-return records from unobserved outgoing control packets.
- Connected the analysis to prior RCAP interface and goalkeeper-behavior investigations; documented configuration and observation limits for follow-up diagnosis.

Analytical projects and experience continue on page 2.

Analytical projects

Bitcoin Volatility and Forecasting

Collaborative academic project | Methodology review and interpretation

- Contributed analytical design and review across historical dataset selection, data cleaning, return and volatility series construction, and evaluation outputs.
- Reviewed methodology involving GARCH(1,1) conditional-volatility estimation and neural-network forecasting. Focused on methodological assumptions and interpretation of evaluation outputs.

Research Data Transformation and Validation

UAV research workflows | Python, ROS, experiment records

- Reviewed trajectory-conversion checks that compared reference and transformed representations, including geometry, timing and downstream behavior.
- Organized reproducible experiment and validation records. Distinguished a runnable pipeline from a valid comparison, and unavailable metrics from measured zeros.

Remote Mine Monitoring and Data Management

Undergraduate capstone | System and data-flow context

- Project scope covered sensor-data acquisition, remote transmission, MySQL storage and host-computer status display.

Experience

PhD Researcher | Universiti Malaya / Roboedge Laboratory | 2026-Present

- Develop and review ROS/Gazebo simulation workflows; investigate dependencies, configuration and runtime behavior. Organize technical requirements, issue descriptions, experiment records and validation steps.

Manufacturing and R&D Support | Family Businesses | 2021-2025

Suzhou Xili Textile / Suzhou Sushanjin, China

- Supported selected material and supplier discussions during early business development; assisted with exhibition materials and delivery preparation, with exposure to production, packaging and transport workflows.

Education and languages

PhD in Computer Science, in progress | Universiti Malaya | 2026-Present

M.Eng. in Mechanical Engineering | Xinjiang University | 2016-2019

B.Eng. in Mechanical Engineering | Xinjiang University | 2012-2016

Mandarin Chinese - native | English - professional working proficiency